Team Name : **SYNTAX ERROR**

Team Leader Name : SOGALAPALLI MANJUNATH REDDY

Problem Statement : Route Resilience: Occlusion-Robust Road Extraction & Graph-Theoretic Criticality Analysis for Urban Mobility

Team Members

Team Leader:

Name: Sogalapalli Manjunath Reddy
College: Keshav Memorial Institute of Technology

Team Member-1:

Name: Anthati Nikhil
College: Keshav Memorial Institute of Technology

Team Member-2:

Name: Gaurav Pendyal
College: Keshav Memorial Institute of Technology

Team Member-3:

Name: Perma Kushal
College: Keshav Memorial Institute of Technology

03 OPPORTUNITY / USP

THE PROBLEM WE SOLVE. THE IMPACT WE DELIVER.

PROJECT GARUDA
Zero-Isolation. Always.

THE REALITY: CHENNAI FLOODS (2023 CASE STUDY)

Extreme rainfall + vulnerable infrastructure =
rapid isolation of critical services

1. HEAVY RAINFALL

Short-duration extreme rainfall overwhelms drainage & roads

2. INFRASTRUCTURE FAILURE

Bridges & roads get cut off, networks fragment

3. HOSPITALS GET ISOLATED

Hospitals & essential services become inaccessible

4. DELAYED RESPONSE, HIGH COST

Rescues delayed, lives & resources at high risk

WHY EXISTING SYSTEMS FALL SHORT

REACTIVE APPROACH
Alerts come after isolation has already occurred

NO CONNECTIVITY INTELLIGENCE
Cannot predict when & where isolation will happen

DATA SILOS
Fragmented data, no fused real-time intelligence

NO ACTIONABLE PRIORITIZATION
Limited ability to optimize resources & routes

THE OPPORTUNITY: PREDICT. PREVENT. PROTECT.

PROJECT GARUDA
ZERO-ISOLATION INTELLIGENCE

GARUDA
Zero-Isolation. Always.

PREDICT – Time-to-Isolation (TTI)
Predicts isolation hours to days in advance with high spatio-temporal

UNDERSTAND – Digital Twin & City Pulse
Real-time digital twin of Chennai to monitor risk, infrastructure health & city pulse

FORESEE – Risk & Percolation Simulation
Simulates flood spread & infrastructure failure to identify vulnerable zones & critical cuts

ACT – Zero-Isolation Engine
Recommends optimal interventions, alternate routes & resource pre-positioning

LEARN – Adaptive Model Learning
Continuously learns from new data & outcomes to improve future predictions

GARUDA transforms fragmented data into predictive, actionable intelligence to ensure Zero-Isolation of lives, infrastructure & essential services.

THE IMPACT

ZERO ISOLATION OF HOSPITALS
No critical services left behind

SAVE LIVES & REDUCE LOSSES
Faster response, lower human & economic cost

RESILIENT & PREPARED CITIES
Data-driven decisions for a climate-resilient future

SCALABLE NATIONWIDE IMPACT
Replicable framework for all Indian cities

**FROM REACTIVE TO PROACTIVE.
FROM ISOLATION TO RESILIENCE.**

PROJECT GARUDA

50 TOTAL INTEGRATED INTELLIGENCE CAPABILITIES

ENGINEERED FOR RESILIENCE INTELLIGENCE • ONE MISSION: ZERO-ISOLATION

ONE MISSION: ZERO-ISOLATION.
No Indian community should become digitally invisible before, during, or after a disaster.

SENSE Observe Continuously → **UNDERSTAND** Build Situational Awareness → **FORESEE** Predict Risks & Impacts → **ACT** Prioritize Actions & Optimize → **LEARN** Adapt & Improve Continuously

12 SIGNATURE INNOVATIONS (What makes GARUDA unique)

01 TIME-TO-ISOLATION (TTI) — Predicts how long specific villages / communities will remain reachable before they become isolated for action.

02 GATEKEEPER NODE RANKING — Identifies & ranks transport critical bridges & roads whose failure can accentuate flood & fragmentation.

03 CITY PULSE SCORE — Real-time resilience score reflecting the city's operational health & transport network status.

04 RESILIENCE DIGITAL TWIN — Live digital replica of the transport network for continuous simulation, impact analysis and planning.

05 NEXT FAILURE PREDICTION — Predicts the infrastructure component most likely to fail next with confidence & risk assessment.

06 DISASTER SCENARIO SIMULATION — Simulates floods, landslides & wind scenarios to understand future impacts & vulnerabilities.

07 INFRASTRUCTURE ACCESSIBILITY — Assesses real-time accessibility of critical infrastructure and evacuation routes across the network.

08 HISTORICAL DISASTER LEARNING — Learns from past disaster patterns to improve prediction accuracy and decision support.

09 INFRASTRUCTURE MEMORY — Maintains long-term structural history using repeated satellite & IoT observations learned from the past.

10 GROUND TRUTH INTEGRATION — Integrates crowd reports, IoT & official sources for accurate & trusted decisions.

11 CRITICAL FACILITY ANALYSIS — Analyzes hospitals, schools, shelters & essential services to ensure continuity of operations.

12 CLOSED-LOOP RESILIENCE OS — Continuously optimizes prepare → act → learn cycle for adaptive resilience.

38 ADVANCED INTELLIGENCE MODULES (High value, sector-specific capabilities)

PERCEPTION & EO INTELLIGENCE (7)
- 01 Optical SAR Fusion
- 02 Multi-Temporal Image Fusion
- 03 Flood Confidence Mapping
- 04 Cloud & Shadow Filtering
- 05 Occlusion Handling
- 06 Change Detection (CD)
- 07 Infrared Hotspot & Fire Detection

7 MODULES

HAZARD & ENVIRONMENT INTELLIGENCE (6)
- 08 Rainfall Nowcasting
- 09 DGM-Based Hazard Mapping
- 10 Hydrodynamic Flood Modeling
- 11 Risk & Vulnerability Alerts
- 12 Wind & Cyclone Path Tracking
- 13 Land Use Change Detection

6 MODULES

RESILIENCE & IMPACT ANALYTICS (7)
- 14 Resilience Indexing
- 15 Critical Facility Analysis
- 16 Population-at-Risk Estimation
- 17 Evacuation Corridor Planning
- 18 Infrastructure Vulnerability
- 19 Transportation Accessibility
- 20 Scenario Impact Evaluation

7 MODULES

DECISION & OPERATION (6)
- 21 Priority Reorder Ranking
- 22 Resource Allocation Support
- 23 Rescue Route Optimization
- 24 Logistics Coordination Engine
- 25 Real-time Incident Support
- 26 Recovery Progress Tracking

6 MODULES

INTERFACE & DELIVERY (6)
- 27 Interactive GIS Layers
- 28 Alert & Notification System
- 29 GeoJSON / API Export
- 30 Mobile & Web Access
- 31 Role-based Dashboards
- 32 Multi-language Support

6 MODULES

ADAPTIVE & LEARNING SYSTEMS (6)
- 33 Continuous Validation
- 34 Adaptive Learning Loop
- 35 Infrastructure Memory Updates
- 36 Truth Validation Engine
- 37 Graph-based Integration
- 38 Crisis Pattern Recognition

6 MODULES

OPEN SOURCE TECHNOLOGIES — Python, PyTorch, GDAL, SNAP, NetworkX, Leaflet

ZERO INFRASTRUCTURE COST — Runs on Free Cloud / Colab / Kaggle GPU

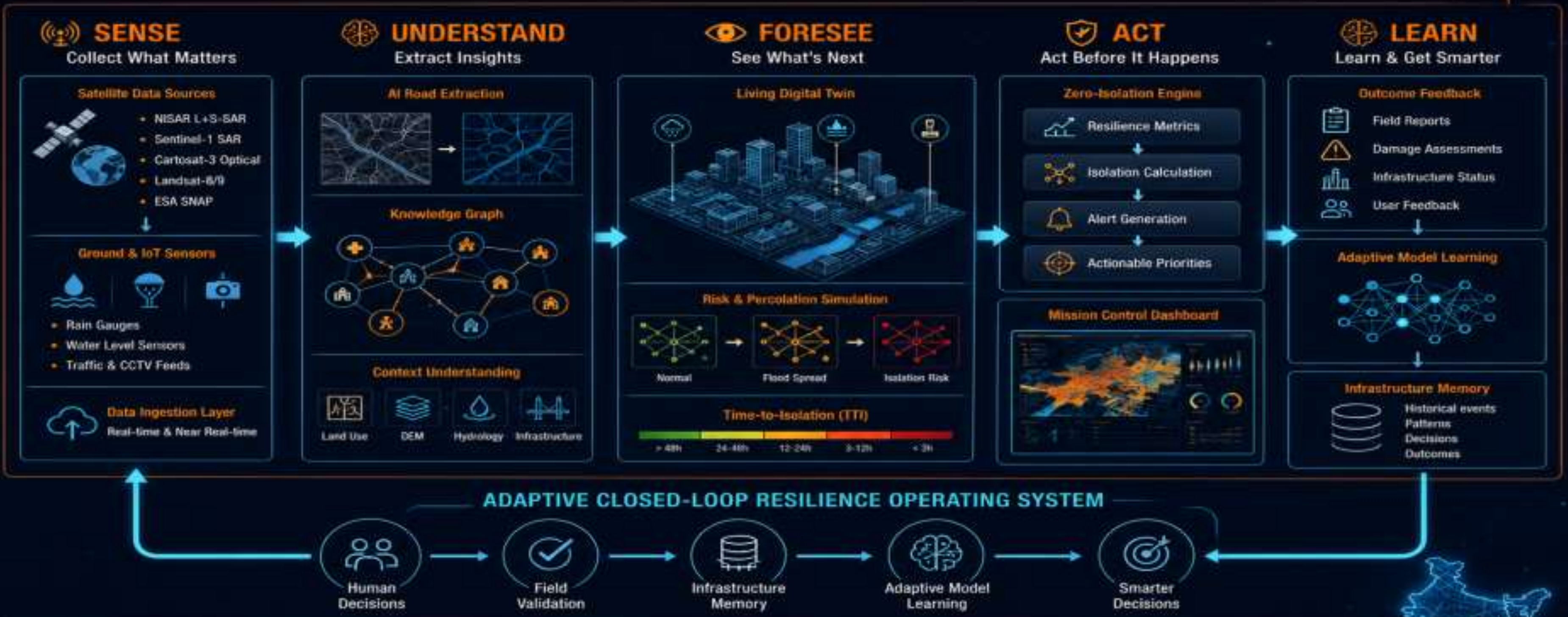
DESIGNED FOR IMPACT — Scalable • Transparent • Actionable for Decision Makers

PURPOSE: ZERO-ISOLATION — Resilient Communities. Connected India. Stronger Nation.

"No Indian community should become digitally invisible — before, during, or after a disaster."

GARUDA turns India's satellites into a disaster-ready nervous system for its cities.

PROJECT GARUDA | SYSTEMS ARCHITECTURE: END-TO-END OPERATIONAL FLOW



ZERO ISOLATION
No critical service left behind

PREDICTIVE POWER
Know isolation hours to days in advance

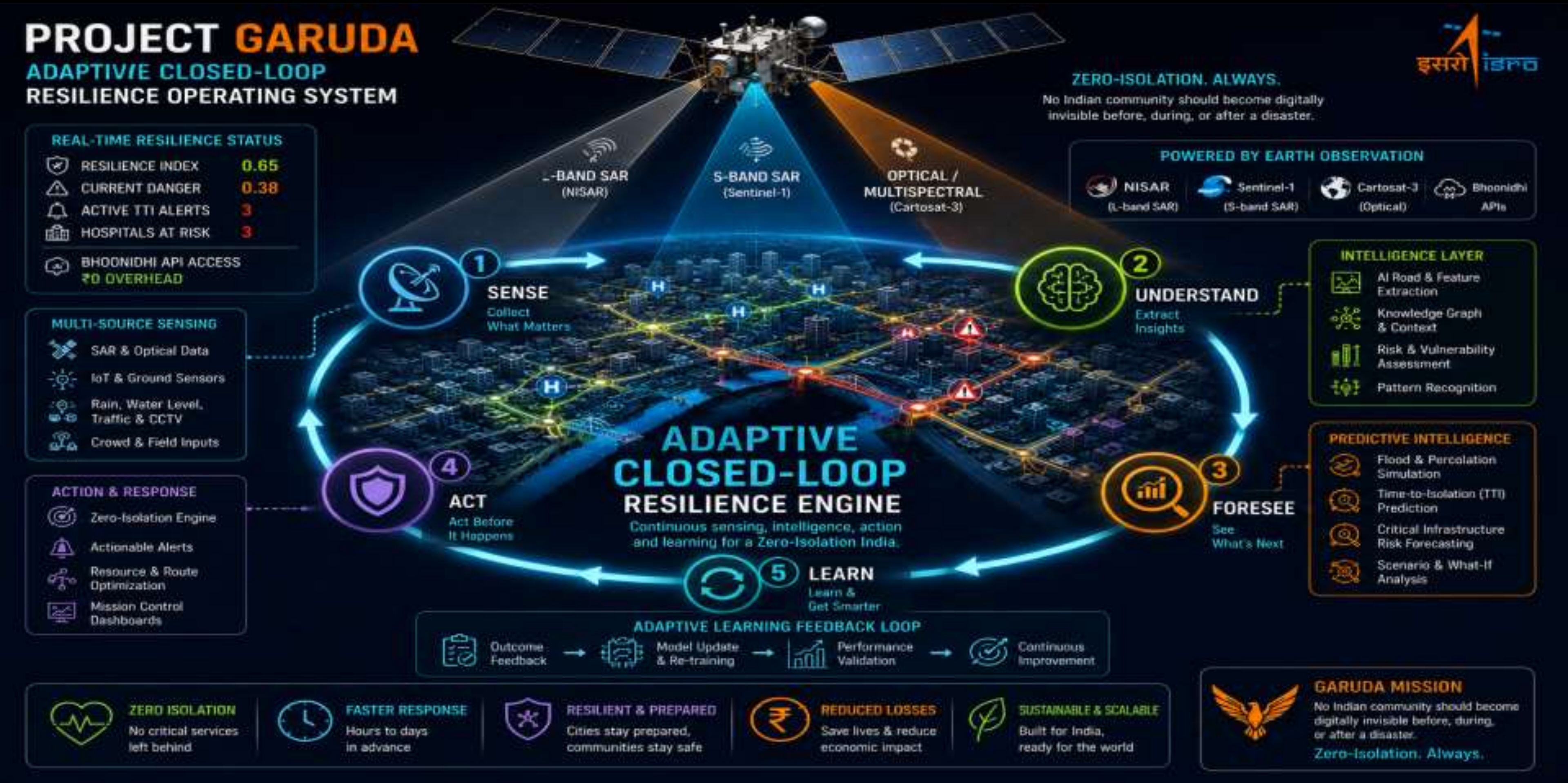
PROACTIVE ACTION
Act before communities become isolated

RESILIENT FUTURE
Data-driven decisions for a safer tomorrow

INDIGENOUS & SCALABLE
Built for India, ready for the world

SENSE. UNDERSTAND. FORESEE. ACT. LEARN. For a Zero-Isolation India.


PROJECT GARUDA
MISSION CONTROL DASHBOARD
Real-time Resilience Intelligence Command Center
28 JUN 2025
11:42:35 IST
SYSTEM OPERATIONAL
Overview
City Pulse
Risk Map
Infrastructure
Alerts
Digital Twin
TTI Monitor
Analytics
Reports
Settings
RESILIENCE INDEX
76 /100
High Resilience
8% vs yesterday
CITY PULSE SCORE
68
Moderate
5% vs yesterday
TIME-TO-ISOLATION (TTI)
42 min
Estimated
15 min vs yesterday
ACTIVE ALERTS
7
High Priority
View All →
AFFECTED POPULATION (EST.)
2.48 L
11% vs yesterday
WEATHER NOW
26°C
Heavy Rain
Wind: 18 km/h
Humidity: 88%
RESILIENCE RISK MAP (Real-time)
RISK LEVEL
Very High
High
Moderate
Low
Very Low
MAP LAYERS
Flood Risk
Infrastructure
Critical Facilities
Transport Network
Water Bodies
Avadi
Poonamallee
Chennai
Velachery
Thiruvanmiyur
Tambaram
REAL-TIME ALERTS & NOTIFICATIONS
View All
Bridge Overtopping Risk
11:41 IST
Adyar Bridge
Water level rising fast. Overtopping possible in 30-45 min.
High Flood Risk Zone
11:39 IST
Okkium Thoraipakkam
Accumulated rainfall high. Flash flood possible.
Traffic Disruption Likely
11:37 IST
GST Road (Chromepet)
Waterlogging detected. Slow traffic expected.
Rainfall Intensity High
11:35 IST
South Chennai
Heavy rainfall continuing. Monitor closely.
7 Active Alerts
View All →
RESILIENCE DIGITAL TWIN
3D CITY MODEL
View 3D
Layers
Simulate
ACTIVE MISSION
CHENNAI FLOOD WATCH
LIVE
Data Source
CARTOSAT-3 + RISAT-1A + GPM
Last Updated
11:42 IST
CRITICAL INFRASTRUCTURE STATUS
View All
Bridges (128)
At Risk: 12
Hospitals (42)
At Risk: 5
Shelters (68)
At Risk: 2
Power Stations (16)
At Risk: 1
At Risk
Warning
Normal
GATEKEEPER NODES (Top 5)
Why Important?
Node | Criticality Score | Impact if Fails | Status
Adyar Bridge | 96 | Very High | At Risk
Kathipara Junction | 91 | Very High | At Risk
Manali New Town | 79 | High | Warning
Velachery Main Road | 72 | High | Warning
Poonamallee High Rd | 68 | Moderate | Normal
View All Gatekeeper Nodes →
AI RECOMMENDED ACTIONS
Powered by GARUDA AI
Monitor Adyar Bridge continuously.
Overtopping probability high in next 45 min.
Priority: HIGH
Pre-position NDRF teams in Okkium area.
High flood risk & evacuation probability.
Priority: HIGH
Implement traffic diversion on GST Road.
Waterlogging likely in next 30-60 min.
Priority: MEDIUM
Alert nearby hospitals & shelters.
Ensure readiness for incoming cases.
Priority: MEDIUM
View Action Plan →
DATA SOURCES
CARTOSAT-3
Optical Imagery
RISAT-1A
SAR Data
GPM IMERG
Rainfall Data
IMD
Weather Data
Census India
Population Data
OpenStreetMap
Base Map
LAST DATA UPDATE
28 JUN 2025, 11:42 IST
GARUDA – Turning India's Satellites into a Disaster-Ready Nervous System for its Cities


PROJECT GARUDA
ADAPTIVIE CLOSED-LOOP
RESILIENCE OPERATING SYSTEM
REAL-TIME RESILIENCE STATUS
RESILIENCE INDEX 0.65
CURRENT DANGER 0.38
ACTIVE TTI ALERTS 3
HOSPITALS AT RISK 3
BHOONIDHI API ACCESS
₹0 OVERHEAD
MULTI-SOURCE SENSING
SAR & Optical Data
IoT & Ground Sensors
Rain, Water Level, Traffic & CCTV
Crowd & Field Inputs
ACTION & RESPONSE
Zero-Isolation Engine
Actionable Alerts
Resource & Route Optimization
Mission Control Dashboards
L-BAND SAR (NISAR)
S-BAND SAR (Sentinel-1)
OPTICAL / MULTISPECTRAL (Cartosat-3)
ZERO-ISOLATION. ALWAYS.
No Indian community should become digitally invisible before, during, or after a disaster.
POWERED BY EARTH OBSERVATION
NISAR (L-band SAR)
Sentinel-1 (S-band SAR)
Cartosat-3 (Optical)
Bhoonidhi APIs
1 SENSE
Collect What Matters
2 UNDERSTAND
Extract Insights
3 FORESEE
See What's Next
4 ACT
Act Before It Happens
5 LEARN
Learn & Get Smarter
ADAPTIVE CLOSED-LOOP RESILIENCE ENGINE
Continuous sensing, intelligence, action and learning for a Zero-Isolation India.
INTELLIGENCE LAYER
AI Road & Feature Extraction
Knowledge Graph & Context
Risk & Vulnerability Assessment
Pattern Recognition
PREDICTIVE INTELLIGENCE
Flood & Percolation Simulation
Time-to-Isolation (TTI) Prediction
Critical Infrastructure Risk Forecasting
Scenario & What-If Analysis
ADAPTIVE LEARNING FEEDBACK LOOP
Outcome Feedback → Model Update & Re-training → Performance Validation → Continuous Improvement
ZERO ISOLATION
No critical services left behind
FASTER RESPONSE
Hours to days in advance
RESILIENT & PREPARED
Cities stay prepared, communities stay safe
REDUCED LOSSES
Save lives & reduce economic impact
SUSTAINABLE & SCALABLE
Built for India, ready for the world
GARUDA MISSION
No Indian community should become digitally invisible before, during, or after a disaster.
Zero-Isolation. Always.

PROJECT GARUDA:
TECHNOLOGY ECOSYSTEM & FRUGAL IMPLEMENTATION MATRIX

1. MACHINE LEARNING CORE

CORE FRAMEWORK

PyTorch

KEY LIBRARIES

- SegFormer / Mask2Former (Vision Transformers)
- Albumentations (Data Augmentation)
- OpenCV (Image Processing)
- NumPy (Numerical Computing)
- Pandas (Data Handling)
- Scikit-learn (ML Utilities)

PURPOSE

AI models for flood mapping, road risk prediction and decision intelligence.

2. GEOSPATIAL & EARTH OBSERVATION DATA

BHOONIDHI API ACCESS
₹0 OVERHEAD

DATA SOURCES & TOOLS

- Cartosat-3 (High Resolution)
- NISAR (L+S-SAR) (All-Weather SAR)
- ESA SNAP (Multi-Temporal Radar)
- GDAL / Rasterio (Geospatial Processing)
- ISRO Bhoonidhi Open APIs
- Sentinel-1 / Landsat Imagery (via Bhoonidhi)
- OpenStreetMap (OSM) (Base Map & Roads)

PURPOSE

Multi-source EO data for real-time situational awareness, monitoring and analysis.

3. NETWORK TOPOLOGY & SIMULATION CORES

CORE TOOLS

KEY MODULES

- NetworkX (Graph Analytics)
- OSMnx (Road Network Extraction)
- Betweenness Centrality
- Percolation Failure Modeling
- Combinatorial Optimization
- Greedy Search Algorithms
- StaMPS InSAR (Deformation Analysis)
- Time-Series Trend Analysis

PURPOSE

Understand network vulnerabilities, simulate failures and identify critical infrastructure risk.

4. FRONTEND, DATA & MANAGEMENT INTERFACES

MISSION CONTROL UI

TECHNOLOGIES

- FastAPI (Asynchronous Backend)
- Streamlit (Dashboards)
- PostGIS (Spatial Database)
- NDEM (Interoperable Formatting)
- React.js (UI Components)
- Leaflet.js (Interactive Maps)
- GitHub (Version Control & Collaboration)

PURPOSE

Seamless data management, visualization and mission decision support.

TOTAL FOUNDATIONAL SETUP OVERHEAD: ₹0 (NO ADDITIONAL INFRASTRUCTURE COST)

RESOURCE SECTOR	FINANCIAL BURDEN	UNDERLYING SYSTEM REASONING
SATELLITE DATA ASSETS Earth Observation Data Access	₹0 ZERO COST	Accessed through ISRO Bhoonidhi and Copernicus open-data services.
COMPUTE OPERATIONS Training & Inference Resources	₹0 ZERO COST	System training optimized on free tier Google Colab and Kaggle Notebook GPU resources.
SOFTWARE LICENSING Open-Source Ecosystem	₹0 ZERO COST	Engineered entirely using open-source packages and frameworks.
ENGINEERING CAPITAL Human Expertise & Development	₹0 ZERO COST	Developed by a four-member interdisciplinary student team using existing institutional computing resources.

Prototype developed entirely using publicly accessible datasets, open-source software, and existing academic computing resources.
Operational deployment can leverage existing ISRO/NRSC infrastructure.

OPEN DATA
ISRO Bhoonidhi, Copernicus

OPEN SOURCE
No licensing cost, maximum flexibility

EXISTING ACADEMIC RESOURCES
Institutions, labs & cloud platforms

NO ADDITIONAL PROTOTYPE COST
100% cost-efficient development

ZERO INFRASTRUCTURE INVESTMENT. MAXIMUM IMPACT. BUILT FOR INDIA.

BHARATIYA ANTARIKSH HACKATHON 2026

THANK YOU